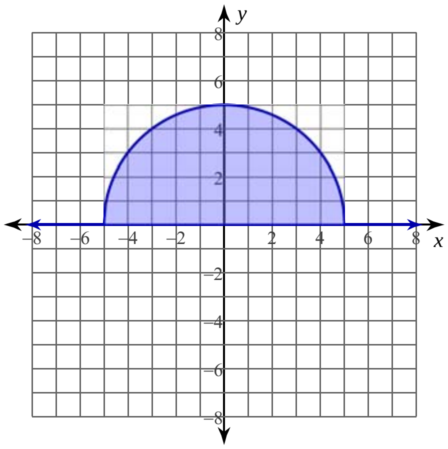
Kuta Software - Infinite Calculus Name_______________________________

Volume of Solids with Known Cross Sections

Date__________________ Period____

For each problem, find the volume of the specified solid. A graph representing the base is provided.

1) The base of a solid is the region enclosed by the semicircle $y = \sqrt{25 - x^2}$ and the x-axis. Cross-sections perpendicular to the x-axis are squares.



2) The base of a solid is the region enclosed by $y = -\frac{x^2}{9} + 4$ and $y = 0$. Cross-sections perpendicular to the x-axis are rectangles with heights twice that of the side in the xy-plane.